STATS & METHODOLOGY

Simple Estimated RAPTOR Projections and Aging Curves



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Here are the particulars for the Estimated RAPTOR version of the Simple Projection System...

The following parameters detail how much weight should be placed on each of the previous three seasons' RAPTOR ratings (multiplied by the player's minutes those seasons) along with a regression-to-the-mean term:

Estimated RAPTOR Simple Projection System weights/values

Parameters for making Estimated RAPTOR forecasts using the Simple Projection System framework

Category	Offense	Defense
Year-1 weight	5.985	5.552
Year-2 weight	2.508	2.542
Year-3 weight	1.508	1.906
Regression to the mean values		
Minutes	2344.144	2757.664
Target	−0.601	−0.121

Table: Neil Paine • Source: Neil's Substack • Get the data • Created with Datawrapper

As an example, if a player was a +3 on offense last year in 2,000 minutes, +2 in 1,500 minutes the year before that and +1 in 1,000 minutes the year before *that*, we'd project his offensive RAPTOR next season to be:

> (3*2000*5.985 + 2*1500*2.508 + 1*1000*1.508 −0.601*2344.144)/(2000*5.985 + 1500*2.508 + 1000*1.508 +2344.144) = 2.2

One other important note is that, while we simplified the example above to use the raw RAPTOR values from previous seasons, the actual projections should age-adjust each previous season's ratings to bring them to their equivalent for the player's age the following season. Here are those aging curves, represented as the expected year-over-year change in RAPTOR rating on either offense or defense based on the age the player will be in the season being projected:

Estimated RAPTOR aging curves

Projected year-over-year change in Offensive and Defensi
Simple Projection System

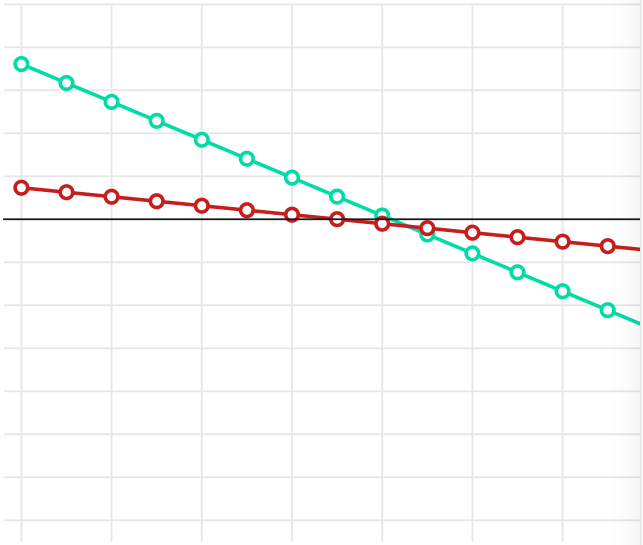


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The final aspect of these simple projecti
most important factors in performance for an NBA rookie are his draft position and his rookie-year age. Here are the parameters for a regression-based estimate of offensive and defensive RAPTOR for players as rookies:

Estimated RAPTOR Simple Projection System parameters for rookies

Category	Offense	Defense
Intercept	−3.474	−0.165
Age coefficient	0.162	0.001
ln(pick) coefficient	−0.564	−0.103

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